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Technical paper

Rescheduling human-robot collaboration tasks under dynamic disassembly scenarios: An MLLM-KG collaboratively enabled approach

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A B S T R A C T

During product recycling, the uncertainty of the degradation level of end-of-life products leads to dynamic conditions such as component corrosion and damage during the disassembly process. Therefore, enhancing the robot's perception of disassembly scenarios and matching historical disassembly experiences is crucial for task rescheduling in human-robot collaborative disassembly (HRCD) under dynamic conditions. To address this, this paper proposes a dynamic task rescheduling method for human-robot collaborative disassembly, empowered by the synergy of Knowledge Graph (KG) and Multimodal Large Language Model (MLLM). Leveraging a Mark-Aware image preprocessing module and prompt-based scene understanding, the physical characteristics and occlusion relationships of disassembly targets are extracted. The concept of affordance is introduced, and an Affordance KG is constructed to recommend disassembly actions based on the physical features of objects in the scene. A task allocation standard for human-robot collaboration is designed, which, combined with depth and human factor information from mixed reality scenarios, enables dynamic task rescheduling and reconstruction of the entire human-robot collaborative disassembly process. The proposed method is validated through a case study on human-robot collaborative disassembly of end-of-life automotive lithium-ion batteries. Experimental results demonstrate that the method exhibits strong robustness and generalizability in dynamic disassembly scenarios, accurately identifying the physical features of components and recommending appropriate disassembly actions under conditions such as component corrosion, damage, and tool unavailability, thus achieving effective task rescheduling.

1. Introduction

Human-Robot Collaboration (HRC) has been a rapidly developing research field in recent years, aiming to address the limitations faced by robots in dynamic manufacturing environments [1].Particularly in the human-robot collaboration disassembly process (HRCD), HRCD not only transfers repetitive and labor-intensive tasks to robots to alleviate the physical burden on humans but also emphasizes the synergy between human intelligence and robotic operational skills [2,3]. However, in dynamic disassembly scenarios, there is often significant uncertainty regarding the degree of product deterioration and the on-site working conditions. For example, corrosion or damage to disassembly objects may complicate component identification, while deformation of parts can hinder the proper use of tools. To ensure the smooth execution of collaborative operations, robots need to fully perceive the current environment and human intentions, and reschedule tasks based on dynamic changes, thereby rescheduling the entire collaborative task [4,5].

Scenario perception methods are a prerequisite for HRCD, directly influencing the robot's reasoning of current task steps and subsequent task rescheduling [6]. Existing scenario perception methods primarily focus on human action recognition and scene graph generation. Specifically, human action recognition [2–4], aims to identify human intentions by analyzing human movements and object information in the scene (such as object type or affordance) to facilitate the understanding of collaborative tasks. Scene graph generation, on the other hand, aims to understand the potential relationships between objects in the scene by building relational graphs, thereby acquiring semantic knowledge of the environment [5–7].

However, there are two issues with existing perception methods when performing task rescheduling in dynamic HRCD scenarios. On the one hand, current scenario perception methods struggle to effectively convey human intentions to robots in the absence of human actions, particularly when predefined tasks are interrupted due to dynamic working conditions, thereby hindering smooth task collaboration [8,9]. On the other hand, despite being trained on large product datasets,

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Abbreviations and nomenclature

HRCD	Human-Robot Collaborative Disassembly
MLLM	Multimodal Large Language Model
KG	Knowledge Graph
MR-HMD	Mixed Reality-Head Mounted Display
AFKG	Affordance Knowledge Graph
P	Part nodes in *AFKG*
F	Feature nodes in *AFKG*
Active	Active component nodes in *AFKG*
Passive	Passive component nodes in *AFKG*
Color	Color nodes in *AFKG*
Shape	Shape nodes in *AFKG*
Material	Material nodes in *AFKG*
D	Disassemble action nodes in *AFKG*
A	Affordance nodes in *AFKG*
T	Tool nodes in *AFKG*
Ac	Automation capability nodes in *AFKG*
RT	Robot disassembly time nodes in *AFKG*
HT	Human disassembly time nodes in *AFKG*
R	Relationships in *AFKG*
belong	Relations connecting *P* or *D* to *A*, indicating the elements in affordance
is	Relations connecting *P* to *F*, indicating the features of parts
after	Relations connecting *Passive* to *Active*, indicating the disassemble sequence among parts
support	Relations connecting *D* to *Ac*, indicating the automation capability of the actions
use	Relations connecting *D* to *T*, indicating the tools used in the actions
need	Relations connecting *T* to *RT* or *HT*, indicating the time consumed with specific tools.
Sg	Scene Graph
T^o	Large Language Model output text
T^i	Large Language Model input text
E	The relational information in T^o
NA	Necessity to automate the corresponding disassembly operation
TAA	Technical ability of a disassembly process to be automated
TETR	Tool and Environment-based Task Rescheduling
DTD	Disassembly task data set

existing methods still face difficulties in handling the changes in product appearance caused by scrapping. Additionally, extracting scene graphs from the internal structure of scrapped products poses further challenges [10,11].

In recent years, Multimodal Large Language Model (MLLM) has garnered widespread attention from the research community due to their powerful image understanding capabilities and convenient text interaction features. In HRCD scenarios, the involved multimodal data includes textual information from historical records, ergonomic data from MR-HMD, as well as image and depth information [12,13]. By collaboratively integrating with Knowledge Graph (KG), MLLM can effectively combine these multimodal datasets within specialized domains, enabling more accurate extraction of human intentions and enhancing the understanding and application of semantic relationships.

Based on this, this paper proposes an approach that collaboratively empowers HRCD task rescheduling in dynamic scenarios by integrating MLLM and KG. First, for the problem of human intent recognition in the absence of explicit human action involvement, we propose a scene understanding method based on MLLM. This method aims to construct a semantic scene graph by leveraging the interaction between the gaze point and the MLLM, combined with on-site 3D image information. Furthermore, for the problem of recognizing unseen components in dynamic disassembly scenarios, an affordance-based knowledge graph (AFKG) is used to query similar affordance scenarios, leveraging the scene graph to match similar disassembly environments and provide disassembly step information. Lastly, a task rescheduling approach is designed based on tool and environmental changes, dynamically reallocating predefined tasks through the depth information of on-site tools and the surrounding environment to achieve overall task rescheduling.

The rest of this paper is organized as follows: Section 2 reviews the relevant literature on scenario perception in the HRC domain, the role of MLLM in semantic understanding, and task allocation in HRCD. Section 3 details the technical aspects of the proposed approach framework. Section 4 presents a case study and experimental results of the proposed method in a human-robot collaborative disassembly scenario for end-of-life automotive lithium-ion batteries. Section 5 discusses the advantages and limitations of the approach, and Section 6 concludes the paper, highlighting future research directions.

2. Related work

2.1. HRC scenario perception

The HRC process includes multiple interrelated modules such as robot perception, recognition, prediction, and execution. First, the perception module is responsible for collecting and processing information from the environment, which forms the basis for recognizing the current intentions and actions of human workers. The recognition module then analyzes this information to accurately assess human behavior and intentions. Subsequently, the prediction module infers the next actions of the human based on the recognition results. Finally, the execution module determines when and how the robot should take action to assist human workers in completing tasks in the most optimal way [8].

This section of research is divided into two categories: scene perception based on human action recognition and scene perception based on scene graph generation. The former extracts human intent and determines task steps by detecting skeletal joint positions, hand movements, and the relative positions with tools and components. This approach is mainly applicable to human-dominated scenarios, aiming to ensure that robots can provide real-time assistance or plan optimal paths to ensure personnel safety. The latter focuses on perceiving object categories in the scene and their relative positional relationships by generating scene graphs. Unlike the former, scene perception methods based on scene graph generation do not rely on participants and can function without human involvement. This method emphasizes the objects and their interrelationships within the entire scene, providing a graph-based representation to enable robots to execute tasks in more complex, multi-participant environments. Specific representative studies are shown in Table 1.

Scene perception techniques based on human action recognition are highly dependent on the presence of human participants. In the absence of participants, such methods cannot be applied to scene analysis, making it difficult to provide effective perception and support in scenarios that involve only the environment and objects. Perception techniques that rely on scene graph generation require a predefined knowledge base to recognize object categories, which typically means that a large amount of training data is needed to accumulate relevant knowledge.

Table 1
Related works on HRC scenario perception.

Classification	Representative Works	Key Techniques	Objectives
Scene perception based on human action recognition	[3,4]	CNN、VMM	Recognize human intentions and predict human actions
	[14]	GCN	
	[15-18]	Dynamic descriptors、Self-attention	Detect potential collisions between humans and robots and plan safe paths
Scene perception based on scene graph generation	[19]	IT2P	Resolve ambiguities in instructions to more accurately locate and pick up target objects.
	[10]	GCN	Improve active planning capabilities in human-machine collaboration to achieve more efficient industrial task execution.
	[11]	Multimodal feature encoder	Improving visual relationship representation in human-robot collaborative assembly to more accurately describe abstract visual relationships and relative spatial features.

2.2. Semantic understanding based on MLLM

MLLM is considered one of the most significant breakthroughs in the field of artificial intelligence in recent years, due to their powerful text understanding and generalization capabilities. However, MLLM has limitations in generating correct text and face difficulties in semantic understanding within specific professional domains. Therefore, research related to semantic understanding based on MLLM aims to enhance their ability to accurately comprehend semantic information in specialized fields and produce correct inference results.

This section of research is divided into three categories: Prompt-based MLLM, Fine-tuned MLLM, and MLLM + External Knowledge. Prompt Engineering methods assist MLLM in organizing input information by designing prompt content, thereby enhancing their semantic understanding capabilities without altering the MLLM structure or pre-trained models. This approach addresses the original MLLM's weak computational ability in complex tasks. Fine-tuning MLLM involves retraining the model on specific semantic context tasks, enabling it to recognize specialized entities in professional fields and output expected semantic information, effectively alleviating the hallucination issues commonly encountered by standard MLLM in specific tasks. The MLLM + External Knowledge approach combines external knowledge bases to avoid the high training costs of original MLLM while further enhancing the model's question-answering capabilities in specialized fields through knowledge retrieval, thereby addressing MLLM hallucination issues to some extent. Specific representative studies are shown in Table 2.

Despite the advantages of MLLM methods as pre-trained models, which reduce training costs to some extent and provide effective means for handling scenes with limited semantic information and relatively stable object states, their application in specialized fields still faces challenges. Prompt-based MLLM methods can effectively address everyday scenarios when processing visual prompts, but they do not sufficiently account for dynamic changes and complex spatial relationships among components in professional domains (such as occlusion issues). Although fine-tuning MLLM can yield expected answers for specific tasks, the high costs and the need for retraining when facing new products undoubtedly increase the investment of time and resources. As for methods that combine MLLM with external knowledge, their performance in HRC scenarios is not ideal. This is primarily due to the difficulty of capturing dynamic changes in obsolete product information with these external knowledge bases, as well as the alignment issues faced by un-tuned MLLM when understanding and applying specialized terminology from the knowledge base.

2.3. HRCD task allocation

HRCD task allocation refers to determining the optimal execution roles for tasks based on various factors such as personnel capabilities, robot functions, and working hours. The core purpose of this process is to optimize human-robot collaboration efficiency and ensure the smooth completion of tasks. Through reasonable task allocation, the respective advantages of humans and robots can be effectively utilized, achieving efficient and safe collaboration.

Current research on HRCD task allocation mainly focuses on three aspects: evaluation criteria, models, and human-robot collaborative optimization. Evaluation criteria emphasize task allocation based on standards such as efficiency, task completion quality, safety, and success rates. Research typically designs specific evaluation metrics or indicator systems to optimize task allocation strategies. These methods iteratively assess and optimize allocation decisions for human-robot collaboration. Models primarily refer to techniques like graph embedding, reinforcement learning, and machine learning to create learnable task allocation models. These models predict optimal solutions for task allocation based on historical or simulated data, often considering task complexity and the complementary capabilities of humans and robots to generate dynamic task allocation strategies. Human-robot collaborative optimization emphasizes the cooperative work between humans and robots during task execution. Through real-time monitoring and optimization, task allocation is dynamically adjusted to adapt to changing environments or task demands. This approach focuses on information feedback and adjustments during task execution to ensure that task allocation is

Table 2
Related work on MLLM in semantic understanding.

Classification	Representative Works	Key Techniques	Objectives
Prompt-based MLLM	[17]	Visual prompting, segmentation models	Unleashing MLLM's visual fundamental capabilities
	[18]	YOLOv7, CLIP	Provide thorough zero-shot UAV scene literary text descriptions.
	[3,20]	Thought chain,	Aims to establish an interactive point cloud intelligent analysis system to provide decision-making support for the construction of urban ecological civilization
Fine-tuned MLLM	[21]	Region prompts,	Provide the coordinates of the mentioned objects in the thought chain and compare the similarity of the areas pointed by the user.
	[22,23]	Segmentation, pixels prompts	Semantic-mask alignment to achieve pixel-level question answering
MLLM + External Knowledge	[24]	Knowledge Graph, link prediction	Determine whether there is a relationship between given skills based on the KG.
	[25]	BERT	Incorporate domain knowledge and semantic information into language model-based methods to better understand relevant common-sense knowledge.

Table 3
HRCD task allocation related work.

Classification	Representative Works	Key Techniques	Objectives
Based on evaluation criteria	[26]	disassembly graph、Methods-Time Measurement	Utilizing teardown diagram technology, an easy-to-use standard catalog is provided to evaluate the automation potential of a given teardown step.
	[27]	multi-objective discrete artificial bee colony (MODABC) algorithm	A taxonomy of uncertainty indicators representing the uncertain characteristics of components and selective disassembly and disassembly transitions is established.
model-based	[28]	ST-GCN	Perceive the overall surrounding scene, identify appropriate collaboration patterns, and assign reconfigurable sequences of operations.
	[29]	Knowledge Graph	A temporal subgraph approach is proposed to infer human-robot-action-component relationships and construct HRC KG for self-organized task allocation.
Based on human-robot collaborative optimization	[30]	Disassembly Planning (DSP)	Human adaptability and robot efficiency are considered to generate feasible DPs.
	[31]		A disassembly task planner was developed to sequentially allocate disassembly tasks between human operators and robots, aiming to address potential issues including different orientations of parts to be disassembled and unsafe conditions.
	[32]	discrete Bees algorithm	Considering human fatigue, disassembly tasks are assigned to humans and robots based on their respective characteristics.

optimal in practice. Specific representative studies are shown in Table 3.

While task allocation based on evaluation criteria allows experts to plan disassembly tasks in advance, many conditions may change in dynamic disassembly scenarios, making it difficult for existing allocation strategies to adapt to real situations. Although model-based allocation can optimize based on data, it faces challenges such as high training costs and the need for substantial data support. Research on human-robot collaborative optimization considers dynamic adjustments during task execution but often overlooks the impact of tool changes on human-robot efficiency. Under dynamic disassembly conditions, human-robot collaborative optimization should incorporate mechanisms for real-time response and adaptation to tool changes to ensure continuous improvement in efficiency.

2.4. Summary

In summary, existing scene perception technologies in the field of HRC face dual challenges: on one hand, their reliance on human action recognition limits their application in scenarios without active participation; on the other hand, the dependence on large datasets and high training costs renders these technologies inadequate for addressing the complexities of dynamic specialized fields. Additionally, while the combination of MLLM and KG demonstrates potential for cross-domain applications, the alignment issue between the general linguistic knowledge output by MLLM and the specialized knowledge in KG remains unresolved. Furthermore, in the HRCD domain, the static nature of existing task allocation methods struggles to adapt to real-time changes in dynamic disassembly environments and tools.

3. Methodology

Collaboratively enabled by KG and MLLM, this paper proposes a task rescheduling approach for HRCD scenarios. An AFKG is defined with the assistance of MLLM-based multi-modal data analysis, so as to match similar HRCD scenarios without action participation in an MR environment. It also reallocates human-robot tasks under real-time changes in disassembly environments or tool replacements, thereby realizing overall task rescheduling.

Fig. 1 displays the entire framework implemented in an MR-based HRCD environment, which consists of three components: (1) Understanding HRCD scenarios based on MLLM; (2) Querying similarity affordance scenarios based on AFKG; (3) Tool and Environment-based Task Rescheduling (TETR). First, the MLLM retrieves on-site images, gaze points, and depth information from the MR-HMD, and outputs a target scene graph based on these inputs. Next, using the target scene graph and the depth information obtained from the MR-HMD, similar affordance subgraphs are queried to identify the disassembly actions required for each step of the disassembly process. Finally, based on the recommended disassembly actions and on-site tool information, task rescheduling for HRCD is conducted and further visualized in the MR environment, guiding humans in performing disassembly tasks.

3.1. Understanding human-robot collaboration disassembly scenarios based on MLLM

The key to architectural design lies in leveraging the powerful image description capability of MLLM to construct scene graphs. This part mainly includes two aspects: (1) Mark-Aware image preprocessing, primarily refining scene semantic information using human gaze locations obtained from the MR-HMD; (2) Prompt design for structured semantic understanding, which involves structuring the semantic understanding of preprocessed images in combination with task requests, and allowing the LLM to directly output the scene graph, supporting subsequent tasks. The specific process is shown in Fig. 2.

3.1.1. Mark-Aware image preprocessing

In dynamic HRCD scenarios, factors such as the similarity between components and changes in component shapes introduce uncertainty. Additionally, different disassembly tasks may result in similarly appearing disassembly scenes. Inspired by Set-of-Mark (SoM) [23], we propose the Mark-Aware (*MaA*) method, which enhances input images I into marked images I_m of human interest, thereby guiding the MLLM to focus on locations of human interest and reducing semantic redundancy. At the same time, by combining the content of prompts T^i inputted into the MLLM $\mathscr{F}$, the output is the target text T^o, represented by Eq. 1:

$$T^o = \mathscr{F}\left(MaA(I), T^i\right) \tag{1}$$

First, the SAM[33] is utilized to mask-segment the input image, extracting meaningful areas from the image, ultimately dividing the input image I, with dimensions H×W, into K regions $R = [r_1, \ldots, r_k] \in \{0,1\}^{K\times H\times W}$. Next, mask dimensions are extracted based on the coordinates of the gaze point. This step primarily involves extracting the target mask r_o related to the gaze point and using distance metrics $d(r_o, R_a) \le 1$ to extract adjacent masks R_a. Subsequently, these masks are used to crop the image, resulting in an image that includes the complete target component mask and its neighboring areas. Finally, the cropped image is marked using the SoM[17] model. The target component mask is numbered as 1, serving as the basis for subsequent prompt queries. This process is represented by Eq. 2:

$$I_m = SoM(f_{crop}(I, r_o, R_a)) \tag{2}$$

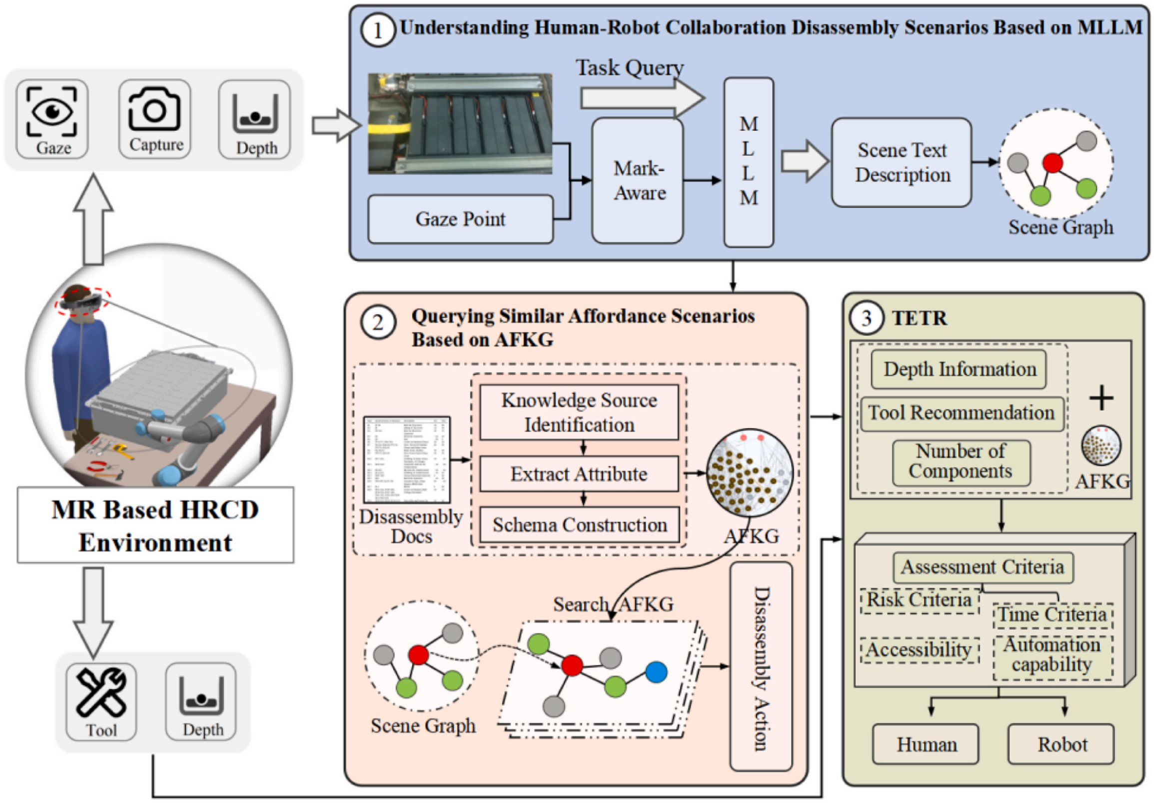

Fig. 1. The framework for KG-MLLM collaboratively enabled HRCD task reconstruction.

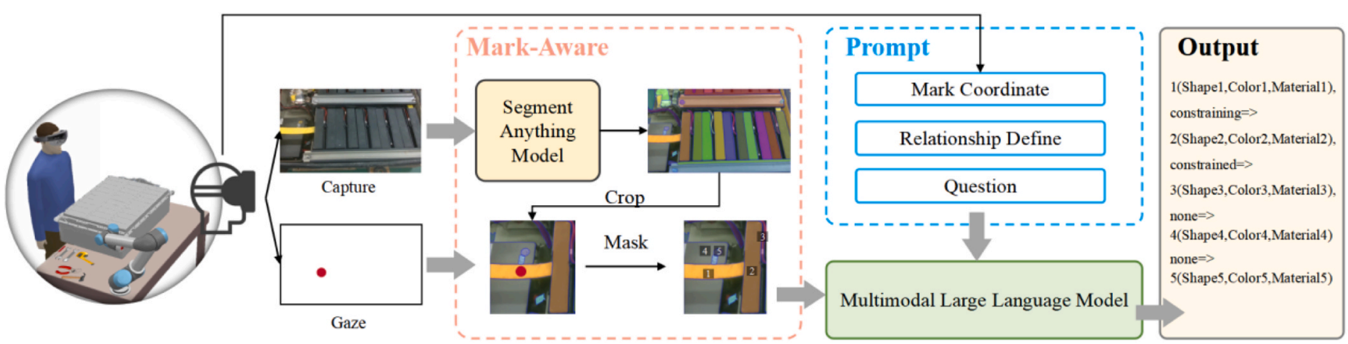

Fig. 2. Illustration of the scene understanding process based on MLLM.

The function f_{crop} represents the cropping function, which crops the image based on the target mask r_o and its adjacent masks R_a. This is because the image encoder compresses the image during the encoding process, which can lead to the loss of some image information, especially in the case of high-resolution images [34]. To avoid the issue of small targets being unrecognized, the regions of interest are pre-cropped. This not only removes irrelevant semantic information but also enhances the visual features of the target object. However, the final marked image I_m contains not only these masks but may also include many smaller surrounding masks. For example, when the target component is a cable and the adjacent component is a battery module, the marking result may also include screws on the battery module and other related components. Therefore, the final cropped image does not ignore critical detail components, and the constraint features between adjacent components are effectively preserved.

After obtaining the marked image I_m, its positional coordinates are input into the MR system, where the spatial coordinates of the marked points are acquired through depth information. This process aims to provide precise location information for the subsequent prompts and to exclude objects that appear to be in contact in the image but are actually on different planes.

After Mark-Aware preprocessing, the resultant image undergoes feature extraction through an image encoder, transforming the image information into embedding representations suitable for LLM processing. These embeddings, along with task requests, are then input into the pretrained large language model. Utilizing its robust multimodal processing capabilities, the model jointly understands and processes both image and textual information.

3.1.2. Prompt design for structured semantic understanding

To ensure that the output of the MLLM conforms to the expected format and semantic information and correctly understands the spatial constraints between objects, it is necessary to integrate the physical features of the objects into easily recognizable colors, shapes, and materials. Taking into account the recognition capabilities of the MLLM, we further define the constraints among these features[34], as designed in

Prompt 1.

Prompt 1 Prompt for scene graph generation

Scene: {Image}
Coordinate: {Spatial Identifier Spatial Coordinates}
Relationship: {Relationship definition}
Instance: {Example}
Question: Please describe the {Color}, {Shape}, and {Material} of each object marked with a spatial identifier based on its Coordinates and Relationship. Also, indicate whether these objects have a Relationship with the object identified by spatial identifier 1. You must choose one from the following categories for {Color}, {Shape}, and {Material}: {Example}
Output: Target object ({Color1}, {Shape1}, {Material1}), {relationship define1} = > ({Color2}, {Shape2}, {Material2})

In Prompt 1, {Image} refers to the image processed using the Mark-Aware method. {Spatial Identifier Spatial Coordinates} is a list that describes the spatial coordinates of each spatial identifier, which helps in assessing spatial relationships. {Relationship definition} refers to the defined relationships between target components and surrounding components during specific tasks, which should be set according to the current task. {Color}, {Shape}, and {Material} are used to describe the features of components in the image, while Instance is used to summarize the appearance features of components commonly found in a specific field (such as battery disassembly).

A scene graph is defined as a graph $Sg = (V, \varepsilon, \{A_k\}_{k=1}^{K})$, with a node set V, an edge set $\varepsilon \subseteq V \times V$, and K attribute sets A_k for $k = 1, \ldots, K$. The scene graph includes the positions of the target objects and their surrounding objects, providing their mutual constraint relationships. Considering the specific output capabilities of the MLLM in this paper, we have determined the output format based on the definition of the scene graph, defining both the target features and their relationships. Ultimately, this information is passed to subsequent steps in the form of triples, thus implementing the entire framework's logic [35,36].

3.2. Querying similar affordance scenarios based on AFKG

The main purpose of this section is to recommend disassembly actions by querying historically similar affordance scenarios through the Target scene graph. This part primarily includes two aspects: (1) construction of the Affordance Knowledge Graph; (2) similarity matching of affordance scenarios based on AFKG.

3.2.1. Object affordance definition

The concept of "object affordance" was first introduced by [37] in 1979 and has recently gained attention in the Human-Robot Collaboration (HRC) field due to its inherent information about the relationships between objects, the environment, and people. For example, the inherent information of pry-ability includes how an object is connected to the environment through snaps or other fastening methods, as well as the actions required by a person to disassemble the object. In addition to containing inherent relationships, object affordance also encapsulates, to some extent, the geometric features of the object and its interlocking components.

Therefore, the definition of object affordance in human-robot collaboration disassembly scenarios in this paper is as follows:

$$Aff_{Sub} = (Sub < Color, Shape, Material >, rel, Obj < Color, Shape, Material >) \tag{3}$$

In this context, *Sub* represents the object described by affordance, and the other elements, the relationship *rel* and the connected component *Obj*, describe the current environment. This illustrates that affordance is not only inherent in the object itself but is also defined by its surrounding environment.

3.2.2. Construction of the affordance knowledge graph

To systematically describe affordance scenarios during the HRCD process and enable querying of historically similar scenes, it is necessary to establish a domain-specific AFKG based on the definition of object affordance Aff_{Sub} and the expert knowledge accumulated in end-of-life products. This AFKG can be divided into eleven node types.

Regarding the structure of parts, this section primarily discusses the definitions of passive components and active components [1]. Standard fasteners, such as bolts, nuts, and screws, as well as clips or covers that must be opened or unfastened, will be labeled as active components. In our definition, active components refer to objects that are directly acted upon by disassembly tools and need to be the first items removed. Passive components, on the other hand, refer to parts that are secured by active components. The removal of these parts generally does not require the use of disassembly tools and can be easily done using a robotic gripper or manually by hand.

Based on the aforementioned data types and their associations, a schema for the KG has been established. Specifically, by analyzing the disassembly step information extracted from process documents and the images of disassembly actions taken on-site, the entities and relational elements shown in the schema were identified. Eqs. 4–7 define the disassembly process affordance ontology model. The specific definitions of node relationships are presented in Abbreviation and Nomenclature.

$$AFKG = \{P \cup F \cup D \cup A \cup T \cup Ac \cup RT \cup HT \cup R\} \tag{4}$$

$$P = \{Active \cup Passive\} \tag{5}$$

$$F = \{Color \cup Shape \cup Material\} \tag{6}$$

$$R = \{belong \cup after \cup is \cup support \cup use \cup need\} \tag{7}$$

As shown in Fig. 3, the schema and an example data set related to a disassembly step of a specific model of retired power battery are displayed. In the human-robot collaboration disassembly scenario, the schema is established based on predefined nodes and edges. Named entity recognition [38], relationship extraction [39], and attribute extraction [40] are performed on the data from the process documents, and nodes and edges are generated with Neo4J [41]. This graph organizes the various affordances in the disassembly steps, providing an informational foundation for subsequent task rescheduling.

3.2.3. Similarity matching of affordance scenarios based on AFKG

After the operations in Section 3.1, the output text of Large Language Model T^o is obtained in the triplet structure of the scene graph, which contains the textual content of target feature m_1 and the constrained and constraining features $\mathscr{M} = [m_2, \ldots, m_k] \in I_m$. This can be represented by Eqs. 8–9.

$$T^o = Sg = m_1(\mathscr{X}), \{E, \mathscr{M}(\mathscr{Y})\} \tag{8}$$

$$E \in (constraining, constrained, none) \tag{9}$$

In Eq. 9, E is determined from constraining relationships, constrained relationships, and no relationships. $\{E, \mathscr{M}(\mathscr{Y})\}$ denotes the relationship and appearance features between the target component and the surrounding components identified. Subsequently, the appearance features χ of the target component are used to perform a similarity query in the AFKG, filtering out nodes from the feature set Y_n that meet the threshold τ. Then, under the affordance nodes ξ connected to these nodes, further queries are made for other component nodes similar to the M component features, ultimately returning the disassembly action nodes A connected to the affordance nodes in this subgraph. The specific querying process is represented by Eq. 10:

$$A = \bigcup_{i=1}^{k} \{A_i | \exists Y_i \in Y_n, Similarity(\mathscr{X}, Y_i) \geq \tau, (Y_i, Y_j) \in \xi, Y_j \sim \mathscr{M}(\mathscr{Y})\} \tag{10}$$

Here, Y_n represents the set of components corresponding to surrounding identifiers, ξ denotes the set of relationships between components, and $Y_j \sim \mathscr{M}(\mathscr{Y})$ indicates the similarity in features between $\mathscr{M}(\mathscr{Y})$ and Y_j. Finally, the disassembly actions A_i associated with these nodes are returned. Fig. 4 illustrates the querying process for similar affordance scenarios, and the detailed process is implemented with Algorithm 1. First, it determines whether the relationships in the triplets T^o are constraining or constrained to decide whether the query nodes are active or passive components. If no nodes similar to the target component m_1 are found, return the triplets in T^o for expert decision on whether to add new relationships to the AFKG. Once similar nodes Y_i are found, search the affordance subgraph for other component nodes, retrieve nodes similar to $\mathscr{Y}$, and return the corresponding disassembly action nodes A. If no similar nodes are found, continue the search from the nodes connected to Y_i and let the expert decide whether to add affordance nodes. If still no match is found, query within the AFKG, and let the expert judge whether to add relationships and affordance nodes. Ultimately, if no similar nodes are found, the expert decides whether to add nodes to the AFKG.

Algorithm 1. Affordance Scenario Similarity Query

```
Input: T^o, AFKG, threshold
Output: action, new AFKG
if max(Similarity(X, Y_n)) < threshold  // Object 1 did not find any similar nodes.
  return Expert_Decision & T^o
else
  Similar_nodes1 ← {}  // Initialize the similar node set for Object 1.
  for each Y ∈ Y_n do
    if Similarity(X, Y) ≥ threshold
      Similar_nodes1.add{Y_i }  // Add similar nodes for Object 1.
    end if
  end for
  if max(Similarity(Y, Y_n)) < threshold  // Object 2 was not found.
    return Expert_Decision  & Y_i, T^o
  else
    Similar_nodes2 ← {}  // Initialize the similar node set for Object 2.
    for each Y ∈ Y_n do
      if Similarity (Y, Y)≥ threshold
      Similar_nodes2 ← Similar_nodes2 ∪ { Y }  // Add similar nodes for Object 2.
      end if
    end for
    related_nodes ← {}  // Initialize the set of nodes with affordance associations.
    for each node1 ∈ Similar_nodes1 do
      for each node2 ∈ Similar_nodes2 do
        if (node1, node2) ∈ Affordance_Relations
          related_nodes ← related_nodes ∪ {(node1, node2)}
        end if
      end for
    end for
    if related_nodes = {}
      return Expert_Decision
    else
      disassembly_actions ← {}
      for each (node1, node2) ∈ related_nodes do
        action ← Affordance_Relations[(node1, node2)]
       disassembly_actions ← disassembly_actions ∪ {Action_Nodes[action]}  // Add disassembly actions.
      end for
      return disassembly_actions
    end if
  end if
```

3.3. Tool and environment-based task rescheduling

Based on the two sections (Sections 3.1 and 3.2), recommended disassembly actions for the current task conditions can be obtained, along with depth information from the MR-HMD. The next step involves

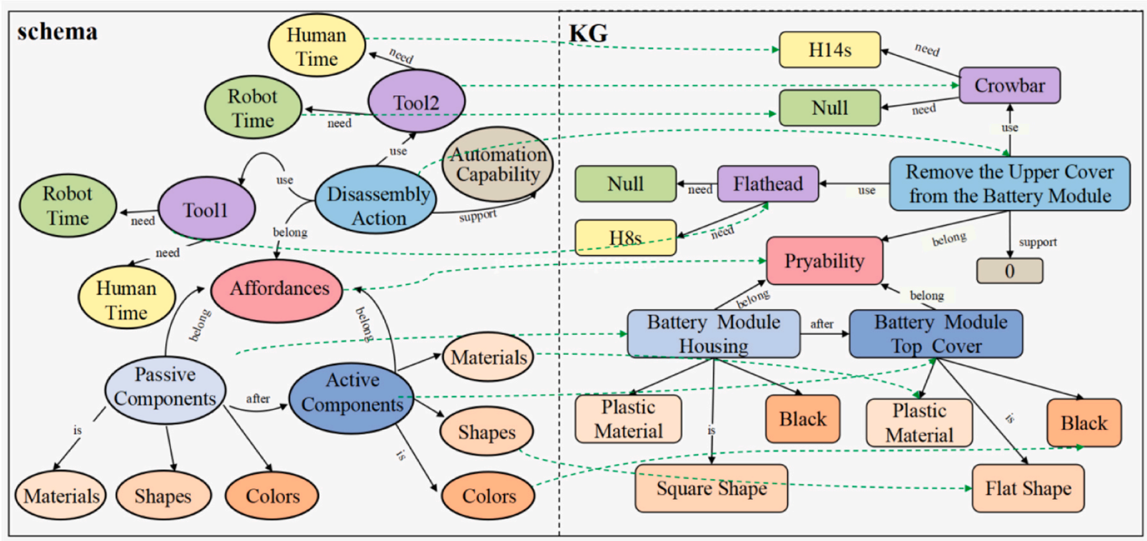

Fig. 3. The schema of AFKG.

evaluating the necessity and ability of robot-assisted disassembly. To facilitate this, a standard catalog has been created to provide criteria for assessing the ability and necessity of robot-led disassembly steps.

Jan[26] and colleagues proposed a catalog containing 10 standards, which we have summarized and simplified into 4 categories primarily used for making decisions based on real-time onsite conditions. Due to the nature of human-robot collaboration, where robots lead but humans assist, the process is not fully automated, allowing some evaluation standards to be simplified. For example, "TAA3 is used to evaluate whether the target is easily detectable visually," which can be adjusted through manual assistance for positional calibration. Additionally, under the NA5 priority aspect, when human intent is involved, the target component is considered the highest priority.

Thus, four assessment criteria for human-robot collaboration disassembly task allocation are proposed here, which are used for the automated planning of task allocation results by robots. Table 4 displays the assessment criteria, including the Necessity to automate the corresponding disassembly operation (NA) and the Technical ability of a disassembly process to be automated (TAA).

For NA, the time required and safety for human operators are the most critical aspects. NA1 takes into account that robots should be considered for disassembly when there are many targets to disassemble, as shown in Table 5. Disassembly time has been approximated based on the Methods-Time Measurement (MTM) system[42].

The criteria for hazards are shown in Table 6. Currently, human-robot collaboration disassembly tasks mainly focus on disassembling from battery packs to battery cells, with hazardous operations primarily involving the disassembly of high-voltage harnesses. Battery packs produced in the industry today are typically painted yellow on high-voltage components as a warning. Therefore, visually, the risk level of disassembly steps can be determined by assessing the color.

Under the premise of meeting the previously mentioned necessity criteria, feasibility (TAA) also needs to be considered. The automation potential of the robot end-effector (TAA1) mainly assesses the variety of tools that the robot can use and the difficulty of the disassembly actions. This evaluation, combined with the automation potential identified in the current research, specifically analyzes the capabilities of the robot end-effector, with details shown in Table 7.

Access (TAA2) describes whether the robot end-effector can reach the target position to perform the disassembly task. For the robot to

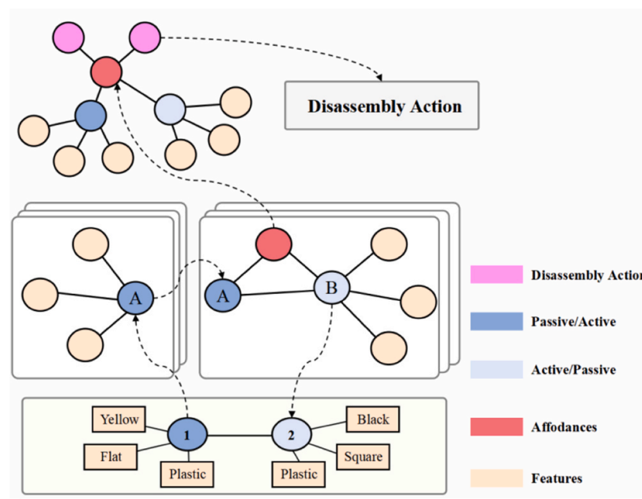

Fig. 4. Query process of similar affordance scenario.

successfully complete the disassembly task, the given end-effector must be able to easily access the active component. This is primarily evaluated through a combination of depth information obtained from MR and the capabilities of the robot end-effector, as detailed in Table 8.

The decision details for task allocation are shown in Algorithm 2. After querying the affordance scenario and obtaining the subgraph, the task allocation decision for the dynamic disassembly scenario is made by combining the depth information obtained from the MR human-robot collaboration disassembly environment, the number of active components, and the on-site tool availability. First, a hazard assessment is conducted based on the component feature descriptions in T^o output by the MLLM. Next, disassembly time is compared for a time evaluation. When NA$>$ 0, it indicates that it is necessary for the robot to execute the disassembly step. Then, reachability and automation potential are assessed. When TAA$=$ 2, it indicates that the robot can feasibly execute the disassembly step.

Algorithm 2. Task Allocation Based on Environment and Tools

Input: AFKG, T^o , num_target_components, depth_info, On-site tool status

Output: Task Executor (Robot or Human)

```
Get disassembly actions' unit times for human and robot from AFKG
Get tool change time for robot from AFKG
Get On-site tool status for human from AFKG
Calculate human_disassembly_time = num_target_components * human_unit_time
Calculate robot_disassembly_time = num_target_components * robot_unit_time
if robot requires tool change
    robot_disassembly_time += tool_change_time
end if
if risk_assessment_from_MLLM > 0
    NA1 = 1
else
    NA1 = 0
end if
if On-site tool status for human is unavailable // the human_disassembly_time should be a large number
    human_disassembly_time = 10E8
end if
if human_disassembly_time < robot_disassembly_time
    NA2 = 0
else
    NA2 = 1
end if
if NA1 + NA2 > 0 // Enough necessity for robot to proceed
    return "Robot"
else
    return "Human"
end if
Get robot end effector tools from AFKG
if tool_recommendation is in robot tools
    TAA1 = 1
else
    TAA1 = 0
end if
Calculate height_diff and radius from depth_info
if height_diff and radius < threshold
    TAA2 = 0
else
    TAA2 = 1
end if
if TAA1 + TAA2 == 2 // Enough technical availability for robot to proceed
    return "Robot"
else
    return "Human"
end if
```

Based on the above method, tasks are reallocated throughout the entire disassembly process to achieve task rescheduling in human-robot collaboration. Disassembly task prompts will be provided through the MR-HMD, ensuring seamless human-robot collaboration. The task rescheduling results produced on-site will be evaluated by experts and stored in the KG, providing a knowledge base for similar affordance disassembly scenarios in the future.

4. Case study and experimental results

To demonstrate the effectiveness and practicality of the proposed MLLM and KG collaboratively enabled HRCD task rescheduling approach, a human-robot collaboration disassembly experiment was conducted in a laboratory environment on a specific model of end-of-life automotive power battery. The experiment was divided into two parts: HRCD case evaluation and a comparative experiment of approaches.

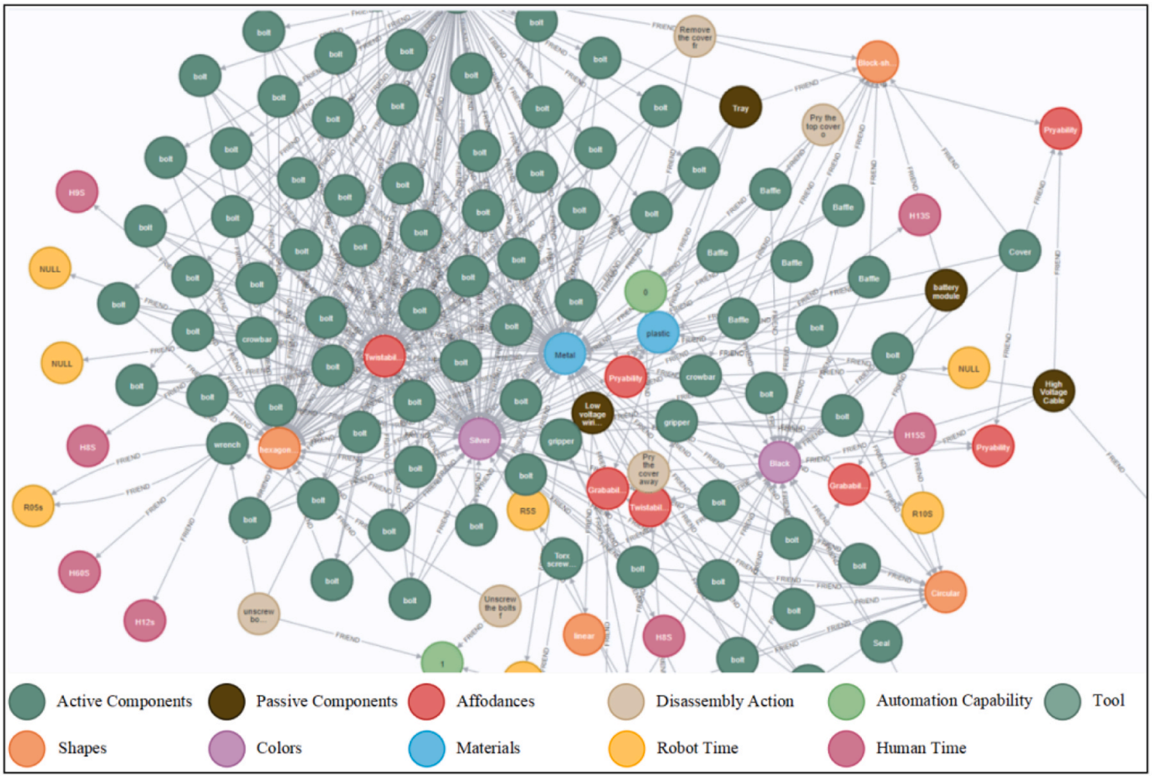

Fig. 5. AFKG diagram.

4.1. Background

4.1.1. Data Collection

As shown in Fig. 6, the data is used to support the requirements of the four steps in the case study. First, the data needed to construct the KG is primarily sourced from Jan's disassembly diagrams [26], Kelly's disassembly research [43], and Chevrolet battery disassembly videos. Second, for a specific model of end-of-life automotive power battery, RGBD images and gaze point coordinates were collected throughout the entire battery disassembly process using HoloLens 2. A total of 400 RGBD images were collected for the disassembly task dataset (DTD) used in the experiment. These images are categorized into four classes based on recognition difficulty, from easy to hard, with 100 images per class.

A) Only the target component and mutually constrained components are present B) Other irrelevant components are present around the target component C) Similar components are present around the target component D) Height differences exist among the components surrounding the target component

4.1.2. Implementation Details

The mentioned MLLM was implemented using LLAVA-v1.6, which, due to its fine-tuning on a large training dataset, demonstrates strong multimodal understanding capabilities and has become a widely used multimodal large language model. Additionally, the experiment utilized the HoloLens 2 MR-HMD, A100 40 G× 8 GPUs, and the UR5 robot. Table 1 presents the prompt feature categories for the MLLM. In this experiment, the hazard classification standard only designates operations involving yellow high-voltage cables as hazardous. Based on the robot's capabilities, grasping and twisting were considered supported operations for the robot. According to the experimental conditions, the robot's end-effector reachability threshold was set to a radius of 5 cm and a height of 12 cm.

4.1.3. KG construction

Since the main objective of this experiment is to disassemble the battery module, only the preceding steps for disassembling the battery module were considered under the priority criteria. Additionally, experts classified and labeled the features of the components and the affordance of the procedures based on the functional structure. Following the steps outlined in Section 3.3.2, perform named entity recognition, relationship extraction and attribute extraction to automatically generate nodes and edges of KG by referring to Pseudocode 1 (readers may refer to [44–46] for more technical details in the algorithm). All edges and nodes were introduced into Neo4J, and the AFKG was constructed using an incremental approach. The result was 153 nodes and 432 edges. Specific details can be found in Table 10.

As shown in Table 11, the original disassembly steps of the experimental object are listed in detail. This human-robot collaboration disassembly experiment primarily involves the process from the battery pack to the battery module. The experiment is considered complete once disassembly reaches the battery module level.

Table 4
Assessment criteria.

Category	Criterion Number	Criterion Description
NA	1	Disassembly time
	2	Danger (High voltage protection)
TAA	1	Automation potential for robotic end effector
	2	Access for end effector

Table 5
Criterion scoring on disassembly time (NA1).

Scoring Value	0	1
Meaning	Indicates that the manual disassembly time is less than the robotic arm disassembly time	Indicates that the manual disassembly time is greater than the robotic arm disassembly time

Table 6
Criterion Scoring on Hazardousness (NA2).

Scoring Value	0	1
Meaning	The disassembly target does not involve high-voltage hazardous components or short-touch operations of positive and negative poles.	The disassembly target involves high-voltage hazardous parts or short-touch operation of positive and negative poles

Table 7
Criterion scoring on the potential of the robot end-effector (TAA1).

Scoring Value	0	1
Meaning	The robot end-effector cannot support this disassembly step.	The robot end-effector supports this disassembly step.

4.2. A case in dynamic scenario

Based on the original battery disassembly steps described earlier, to validate the effectiveness and practicality of the proposed method in dynamic environments for human-robot collaborative disassembly, the following scenarios with frequently changing conditions are highlighted:

Dynamic Scenario 1: The external features of the disassembly object undergo significant changes, and the dissolution of rubber causes bonding between components. In this case, the disassembly priority increases significantly, requiring dynamic adjustment of the step sequence and precise identification of objects after their appearance changes.

Dynamic Scenario 2: The disassembly object is severely corroded, causing components to fall off automatically without the need for disassembly. In this situation, it is necessary to accurately identify the on-site procedures to determine whether this step can be skipped.

Dynamic Scenario 3: Due to dynamic changes in on-site tools, some tools may be missing. It is first necessary to accurately identify the affordance of the scene and detect alternative tools available on-site.

4.2.1. Understanding disassembly scenarios with Mark-Aware MLLM

In this subsection, the interaction process of the MLLM and its output results are presented. Fig. 8 illustrates the image processing steps of the Mark-Aware module in dynamic scenario 1. First, HoloLens captures image information and generates a mask for the entire image using the SAM model (Image 1). Then, the depth sensor of HoloLens filters out distant objects, retaining only the masks of objects within the same plane (Image 2). Next, non-adjacent mask areas are removed to obtain Image 3, and blank areas without masks are cropped out, resulting in Image 4. After labeling the mask areas, the processed image is generated.

After obtaining the processed image, it is combined with the prompt information and input into the MLLM. As shown in Fig. 9, the image illustrates the visualization of the interaction results of the MLLM on the HoloLens interface. Additionally, the figure displays the mixed reality (MR) interface visualizations of the captured images and image processing for dynamic scenario 1 and dynamic scenario 2.

Table 8
Criterion scorings for the access (TAA2).

Scoring Value	0	1
Meaning	The surrounding environment of the target active component does not allow the end-effector to approach.	The surrounding environment of the target active component allows the end-effector to approach.

Table 9
Prompt feature categories.

Id	Shapes	Colors	Materials
1	Circular	Black	Metal
2	Block-shaped	White	plastic
3	linear	Yellow	transparent
4	tubular	Silver	other materials
5	sheet-like	Transparent	
6	hexagonal	Other colors	
7	strip-like		
8	other shapes		

The output results from MLLM demonstrate that the Mark-Aware + MLLM architecture can effectively identify predefined relationships between objects. Furthermore, for objects with multiple color blocks (such as the yellow cable's outer casing and terminals), even if Mark-Aware has segmented and marked them, MLLM can still determine whether these marked objects belong to the same entity and merge them into a single object for relationship assessment. This indicates that the architecture exhibits high accuracy and flexibility when handling complex objects.

4.2.2. Querying similar affordance scenes based on AFKG

In the context of human-machine collaborative disassembly scenarios based on MLLM, three dynamic scenarios have produced the three segments of text information shown above in Fig. 10. To verify the reliability of the affordance query results, we combined these three segments of text information with the AFKG constructed in Section 4.1 to query historically similar disassembly scenarios.

First, by utilizing the features of the target component and the features of its related constrained components in the KG, we queried nodes with similar characteristics and filtered them through the after relationships between nodes, ultimately obtaining similar affordance disassembly scenarios, as shown in Fig. 10(a).

Next, based on the queried affordance nodes, we searched for the associated disassembly actions and their automation capabilities. A value of 0 for automation capability indicates that the laboratory robot is unable to perform the action, while a value of 1 indicates that the robot has the capability to perform the action, as shown in Fig. 10(b).

Finally, based on the obtained disassembly action nodes, we queried the related disassembly tools to determine the disassembly times for both humans and the robot. In this step, the disassembly time includes preparation time, representing the total time required to complete the disassembly of a single component. The yellow node R3s indicates the time required for the robot to complete the disassembly of a single component in that step, while the purple node H12s indicates the time required for a human in the same step, as shown in Fig. 10(c).

4.2.3. Rescheduling tasks in dynamic scenarios

Based on the similar availability query results derived from AFKG mentioned in Section 4.2.2, the automation capability TAA1 and unit disassembly time can be obtained, allowing for the calculation of the corresponding NA1. Subsequently, the tasks can be reallocated to achieve task rescheduling, with the results shown in Table 12. The final expert evaluation indicates the effectiveness of the proposed method.

In dynamic scenario 1, since there is only one component and it is not hazardous, the NA (Necessity for Automation) value is 0. However, due to the adhesive nature of the dissolved sound insulation material, the robotic arm cannot perform the related automated operations, resulting in a TAA (Technical Ability) value of 1. With a total score of less than 2, the allocation result is "human."

In dynamic scenario 2, although there is only one component to be removed, the involvement of hazardous operations gives it an NA (Necessity for Automation) value of 1, indicating that automation is necessary for this step. However, in the analysis of depth information, due to the damage to the cover, the surrounding environment does not

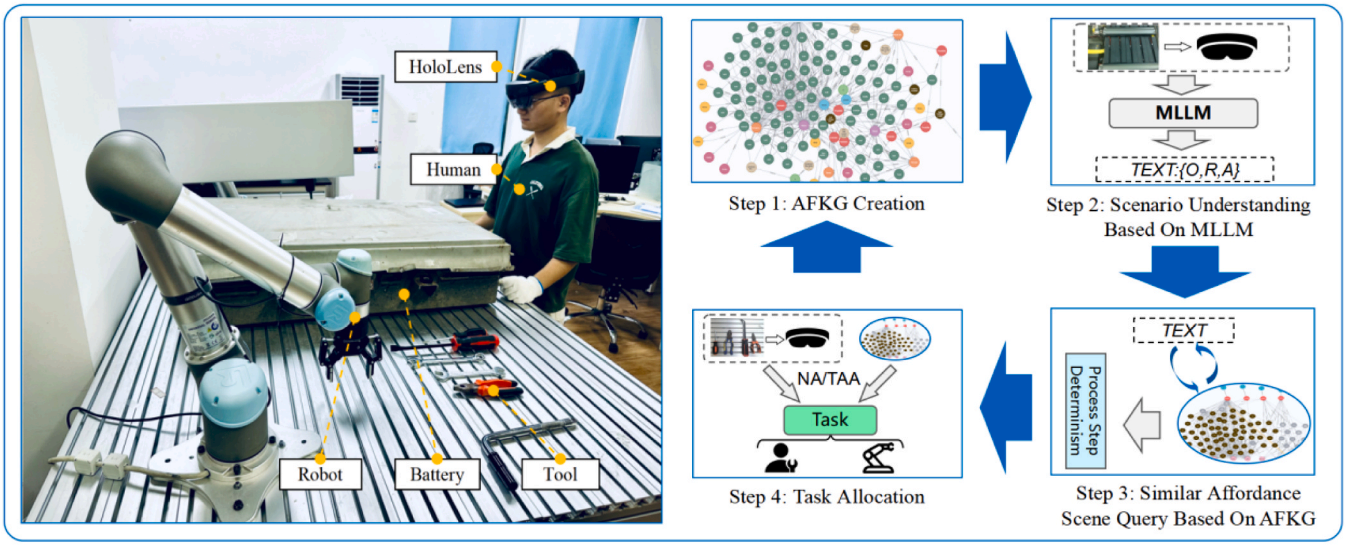

Fig. 6. Experimental case diagram.

meet the conditions for automation, resulting in a TAA (Technical Ability) value of 0, indicating that the robot cannot perform the disassembly operation. Therefore, the allocation result also changes to "human."

In dynamic scenario 3, initially, the number of target components is small and does not involve hazardous operations, so there is no necessity for automation in the disassembly operation. However, due to the absence of a wrench, the waiting time for manual disassembly far exceeds the operation time of the robot, resulting in an NA (Necessity for Automation) value of 1. At the same time, for this step, the robot has the capability for automation and the surrounding environment is accessible, giving a TAA (Technical Ability) value of 2. The total score exceeds 3, indicating that the disassembly task can be allocated to the robot, allowing the disassembly task to proceed normally.

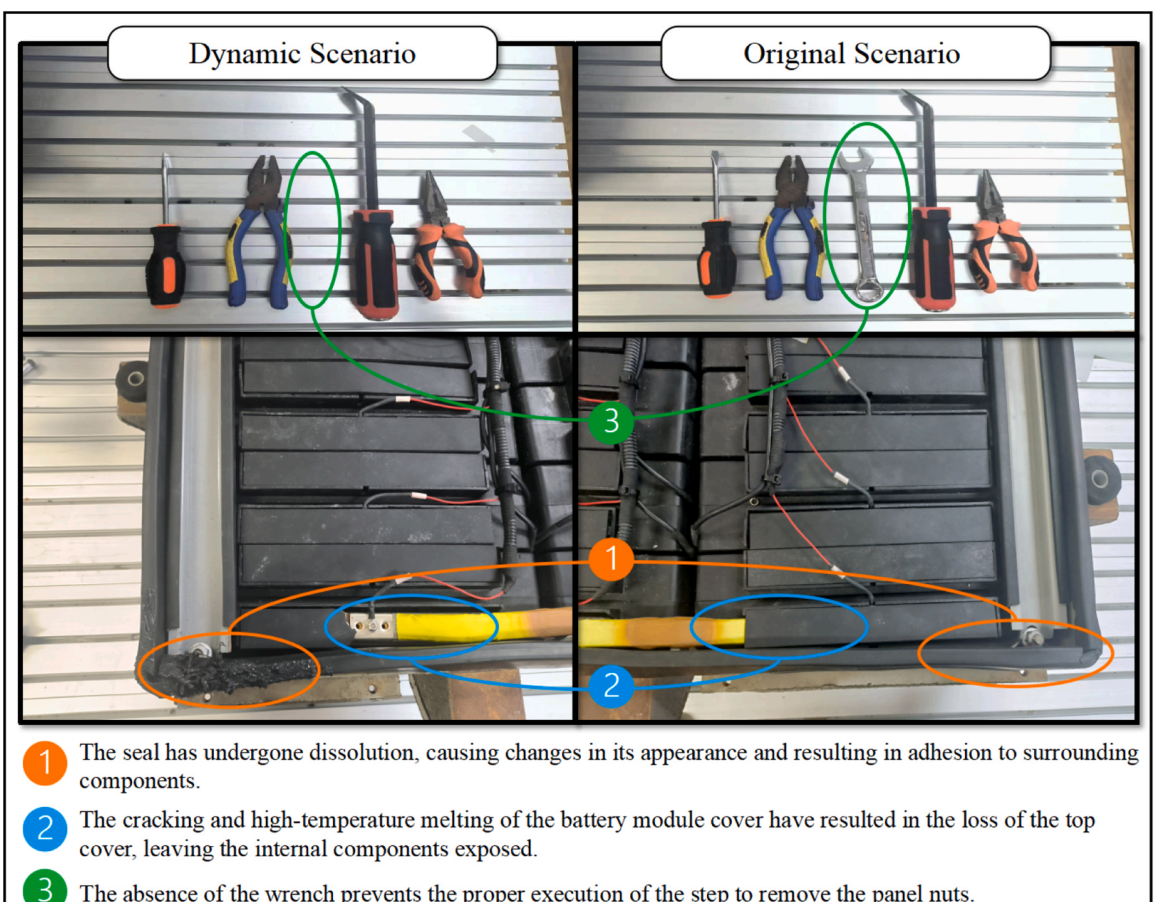

Fig. 7. Dynamic scenario illustrations in the case study.

Table 10
List of node types and their counts in the graph.

Node types	Counts
Active components	78
Passive components	24
Affodances	12
Colors	4
Materials	2
Shapes	6
Disassembly action	8
Human time	10
Robot time	10
Tool	5
Automation capability	2

With the above results, the effectiveness of the proposed MLLM and KG collaboratively enabled HRCD task rescheduling approach is demonstrated. For general practical applications, two perspectives can be further concerned. First, most aging batteries on the market are highly similar to the batteries disassembled in this paper, which primarily consist of battery cells, modules, and battery packs (namely, CMP structure). The physical characteristics of the components are largely identical, and the constraints are mostly the same. Therefore, in the disassembly steps specific to different models of batteries, appropriate adjustments to the robot's end effector's disassembly radius could be easily conducted in the proposed manner to enable effective technology transfer under new HRCD cases. Moreover, for complicated situations where some parts are completely overlapped and cannot be directly observed, they can be identified by leveraging the neighboring observable parts and performing multi-hop relation queries on the AFKG, in which some prior knowledge is collected during the disassembly process and then applied to enhance this accuracy. The final affordance scene matching results will be also sent to frontline workers for further inspection, thereby guaranteeing collaborative safety and ensuring human-robot collaboration accuracy.

4.3. Comparative experiment

4.3.1. Tasks

Section 4.1 describes the data collection process and categories of the 400-image dataset (DTD). It primarily includes four categories involving multiple disassembly steps and covers the three dynamic scenarios mentioned in Section 4.2. In the comparative experiment, results from the 400 images are evaluated. The specific experimental plan involves replacing the scene perception module of MLLM with other MLLM methods, while keeping the AFKG and task allocation TETR modules unchanged, to compare results on DTD. The specific experimental data includes two tables, with Table 13 showing results where only images are provided, and the model needs to correctly predict the positions of all objects in the image and the pairwise relationships with the target objects. Table 14 presents the accuracy of task allocation results after the change in the scene perception module.

4.3.2. Evaluation metrics

Following [50], this paper uses Recall@K (R@K), Accuracy@K (A@K) to measure the recall and accuracy of the MLLM method in scene graph generation. This is done by comparing the top K predicted triples (subject-predicate-object) with the ground truth. In this experiment, the value of K is uniformly set to 1 [17]. The final task allocation results of the entire experiment are measured using Accuracy. All the evaluation metrics are the average values of the output results on the DTD dataset.

Table 13 presents the experimental results of various comparative methods on three DTD datasets. As shown in the table, the MLLM method alone achieves a high recognition accuracy in scenarios with similar parts, but it is accompanied by a relatively low recall rate. This suggests that the MLLM method is more suitable for tasks such as product defect detection in similar part scenarios. In contrast, other methods based on MLLM prompting perform better in terms of recall in disassembly scenarios, but their accuracy does not reach the desired level. This indicates that these methods are more appropriate for target detection and retrieval tasks in disassembly scenarios. The framework proposed in this paper outperforms the other methods in all three scenarios, particularly when recognizing parts with high differentiation. Its scene perception ability is significantly superior to that of the comparative methods, demonstrating that the proposed method is more suitable for identifying part constraint relationships in the scene and extracting the relevant relationships.

Besides, in Fig. 11(a), MLLM frequently misidentifies the irrelevant component a-6, while other MLLM prompting methods tend to analyze components a-1 (rubber) and a-2 (terminal) as two separate entities. In Fig. 11(b), MLLM often overlooks the similar component b-3, whereas the prompting methods overall effectively address this issue. As for Fig. 11(c), all comparison methods incorrectly predicted the relationship

Table 11
Battery disassembly steps.

Sequence	Disassembly Components	Disassembly Actions	Executor	Disassembly Tools
#1	Upper casing screws	Unscrew the 24 bolts on the upper casing	Robot	Phillips screwdriver
#2	Battery outer casing	Pry the upper casing away from the lower tray	Human	Pry bar
#3	Battery module cover	Pry the battery module cover off the battery module	Human	Pry bar
#4	Baffle nuts	Unscrew the nuts from the baffle	Human	Wrench
#5	Yellow cable screws	Unscrew the screws on the yellow cable	Robot	Screwdriver
#6	Yellow high-voltage cable	Remove the yellow cable from the battery module	Robot	Robotic gripper
#7	Busbar screws	Unscrew the screws on the busbar	Robot	Phillips screwdriver
#8	Low-voltage harness screws	Loosen the set screws on the low-voltage harness	Robot	Phillips screwdriver
#9	Baffle	Pry the baffle away from the battery module	Human	Pry bar
#10	Busbar	Pry the busbar away from the module	Human	Pry bar
#11	Seal	Remove the seal from the lower tray	Robot	Robotic gripper
#12	Battery module	Remove the battery module from the lower tray	Robot	Robotic gripper

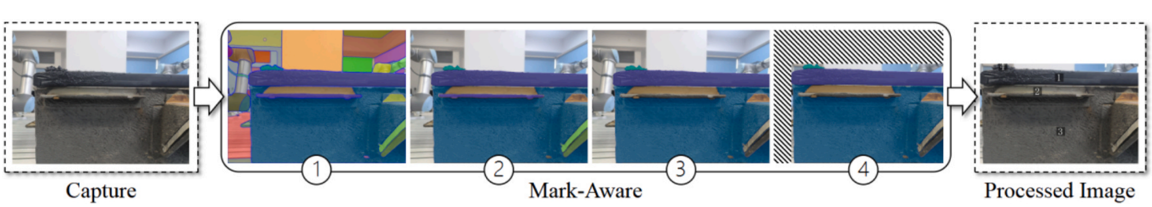

Fig. 8. Flowchart of the Mark-Aware module processing.

Fig. 9. MR interface display.

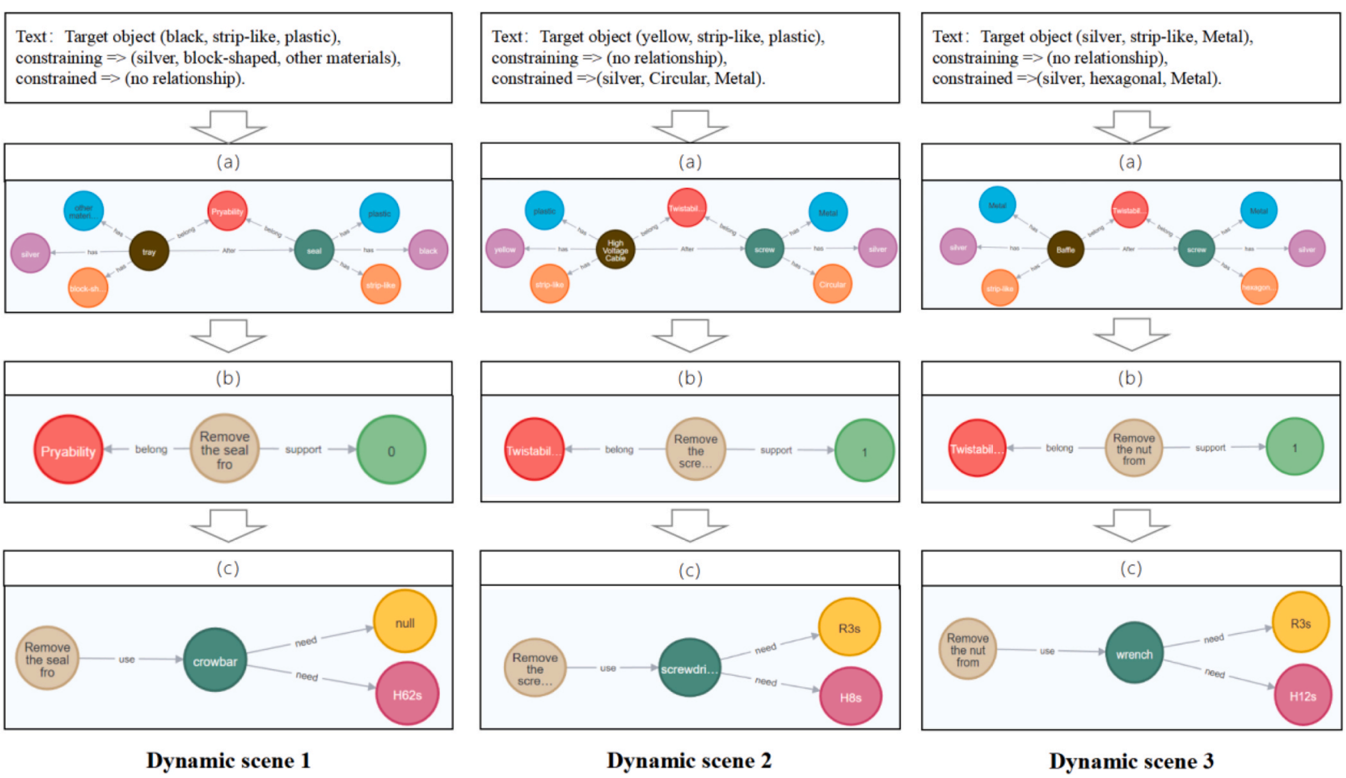

Fig. 10. Display of subgraph query results.

between the target component c-1 and the other tall component c-2.

In the comparison of task allocation accuracy, the proposed framework also outperforms other methods, as detailed in Table 14. Additionally, independent experiments were conducted for the three dynamic scenarios, showcasing the correct task allocation results when all three scenarios occur simultaneously, along with the allocation results from the comparison methods, as shown in Table 15. The data in the second row under "expert experience" indicates that the main changes in task allocation during these three dynamic scenarios are concentrated in steps #4, #5, and #11. The table data shows that the

Table 12
Task rescheduling results in dynamic scenarios.

Dynamic scene	Phenomena	Changed procedures	Initial allocation	NA	TAA	Expert experience	Allocate after changes
1	Dissolution of seal	#11 Remove seal from the lower tray	Robot	0	1	Human	Human
2	Damage to battery module cover	#5 Unscrew the screws from the yellow cable	Robot	1	0	Human	Human
3	Wrench missing	#4 Unscrew the nuts on the baffle	Human	1	2	Robot	Robot

Table 13
Recall and accuracy performance in scene graph generation.

DTD Models	(a) irrelevant parts around (%)		(b) similar parts around (%)		(c) height difference around (%)	
	R@1	A@1	R@1	A@1	R@1	A@1
LLaVA+AFKG+TETR[47]	77.3	74.1	68.3	75.9	62.5	62.8
GLM-4V+AFKG+ TETR[48]	77.4	75.4	71.5	76.6	65.8	63.6
MLLM+SoM+AFKG+TETR[17]	78.6	75.7	79.7	76.8	68.9	65.9
ControlMLLM+AFKG+TETR[49]	78.9	81.0	79.0	75.6	70.5	71.3
Mark-Aware+MLLM+AFKG+TETR	**86.0**	**82.1**	**81.2**	**78.3**	**78.6**	**75.5**

Table 14
Task allocation results on three DTD images.

DTD Models	(a) irrelevant parts around (%)	(b) similar parts around (%)	(c) height difference around (%)
	Accuracy	Accuracy	Accuracy
LLaVA+AFKG+TETR	54.50	57.52	59.63
GLM-4V+AFKG+ TETR	58.32	56.73	60.17
MLLM+SoM+AFKG+TETR	64.73	62.59	68.09
ControlMLLM+AFKG+TETR	66.98	67.54	69.73
Mark-Aware+MLLM+AFKG+TETR	**67.24**	**69.27**	**74.80**

comparison methods not only made allocation errors but also missed the correct distribution results after changes, while the task allocation results of the proposed framework align with the expert experience.

To further evaluate the effectiveness of the proposed framework, two ablation experiments are conducted in Table 16. One experiment removes the AFKG, leaving only the MLLM method for scene perception

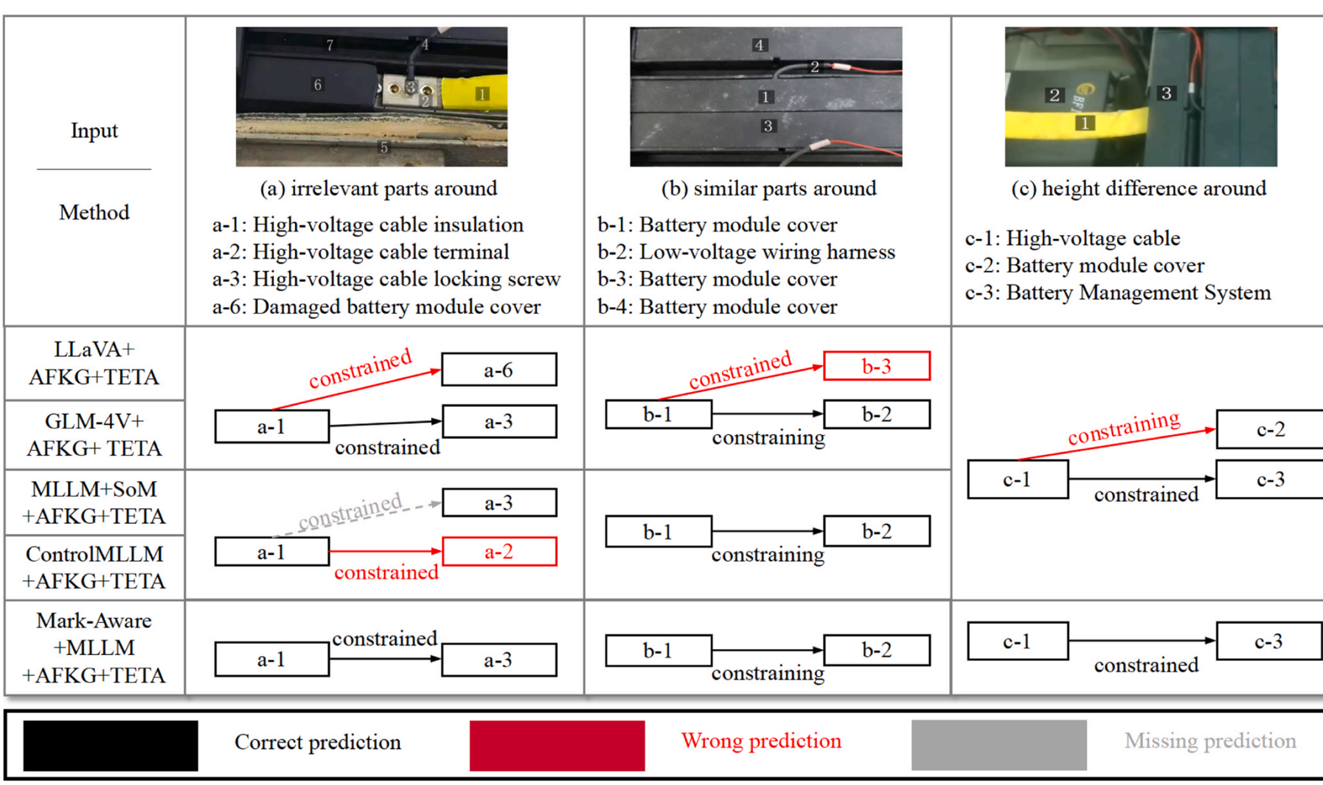

Fig. 11. Outputs of the comparative experiment on DTD.

to determine the necessary disassembly actions and tools for the disassembly task. The task allocation (TETR) is then performed based on the available tools and environment. The other experiment eliminates the TETR method, allowing the MLLM to perform task allocation based on the query results from the AFKG.

The experiments utilize the datasets of the three dynamic scenarios mentioned in the case study from the DTD dataset, measuring the average accuracy of task allocation results for the corresponding steps across the three methods. The proposed framework demonstrates excellent robustness in both dynamic and static scenes compared to other methods.

Table 16
Ablation experiment results.

Process step	Disassembly operation change	Proposed	w/o AFKG	w/o TETR
#4	Due to the missing wrench, the robot unscrews the bezel nut instead	**74.80**	67.73	51.25
#5	Due to the damaged cover, humans tightened the screws instead	**69.27**	62.57	58.72
#11	Due to the dissolved sealing ring, humans pried it loose instead	**67.24**	61.48	56.14
Others	No changes	**75.56**	64.74	59.25

5. Discussions

This paper addresses the uncertainties in component scrap levels and tool variations in dynamic disassembly scenarios by proposing a task rescheduling method based on the collaboration of MLLM and KG. Through experimental cases, the framework's effectiveness and versatility are verified. Compared to existing MLLM prompting methods, the proposed framework has three major advantages in handling dynamic human-machine collaborative disassembly scenarios.

First, as shown in Table 13 and Fig. 11, the proposed framework adapts well to disassembly scenes with complex spatial relationships, such as occlusion, in scene graph generation. In contrast, traditional MLLM-driven frameworks may overlook target identification when there are many similar parts, while other MLLM prompting methods struggle with accurate recognition in the presence of complex colors. This is because, under Mark-Aware prompting, MLLM can filter out distracting background information and capture depth information for better adaptation to complex constraints and occlusions in disassembly scenes. Additionally, compared to traditional intent recognition methods that rely on human actions, the Mark-Aware-empowered MLLM enables human intent recognition without the need for any physical actions. This significantly enhances the flexibility of the robot's selective disassembly.

Second, the ablation study data in Table 16 indicates that the task reconstruction accuracy significantly decreases when the KG is removed. Without the specialized domain knowledge provided by KG, MLLM cannot ensure that its operational judgments based on inherent knowledge are sufficiently accurate. As a result, the accuracy of the step information obtained during task reconstruction drops considerably. Moreover, compared to traditional methods, the KG-empowered MLLM achieves accurate recognition of previously unseen parts in disassembly scenarios, without incurring the high training costs. This demonstrates its outstanding generalization capability and high efficiency.

Finally, the experimental results in Tables 14–16 demonstrate that the proposed framework exhibits superior performance in task reconstruction. In contrast, the comparison methods suffer from inaccuracies in scene semantic understanding, leading to errors in similarity matching within affordance scenes. This results in missing correct affordance node information during task allocation, ultimately affecting the accuracy of task rescheduling. In contrast to traditional methods, the framework proposed in this paper is capable of accurately rescheduling human-robot collaboration tasks even in complex scenarios where there is no physical action involved and previously unseen parts are present in the scene. This significantly enhances both the accuracy and efficiency of task execution.

Despite the aforementioned advantages, the proposed framework still has two limitations. Firstly, this paper focuses solely on the robot's perception capabilities in dynamic scenarios, aiming to assist robots in collaborating with humans to complete tasks. However, the current study does not address the requirements and limitations of robot operations in dynamic scenarios. Future research will build on this to further explore how robots can enhance their collaborative capabilities in complex environments.

Secondly, some disassembly scenarios involve multiple work areas operating in parallel, which involves multiple robots or multiple individuals performing disassembly tasks simultaneously. However, the framework proposed in this paper only considers single-person, single-machine collaborative disassembly scenarios. To solve this, a task management module can be added, or multi-agent task optimization information can be introduced into the KG to better support complex multi-agent collaborative operations.

6. Conclusions

In dynamic disassembly scenarios, uncertainties in component scrap levels and tool variations often lead to the appearance of previously unseen parts. Traditional scene perception methods rely on large training datasets for learning on known components and require human actions to capture human intent, making them difficult to handle disassembly tasks involving unseen parts. To address this challenge, this paper proposes a task rescheduling method based on MLLM + KG for dynamic human-robot collaborative disassembly scenarios, aimed at identifying appropriate tools and rationalizing task allocation in complex disassembly environments. The main contributions can be summarized in three aspects.

Table 15
Task allocation results of comparison methods under three dynamic scenarios occurring simultaneously.

Sequence	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12
Initial allocation	R	H	H	H	R	R	R	R	H	H	R	R
Expert experience	—	—	—	H~R	R~H	—	—	—	—	—	R~H	—
LLaVA+AFKG +TETR	—	—	H~R	H~R	—	—	—	R~H	—	—	—	—
GLM-4V+AFKG + TETR	—	—	H~R	—	—	—	—	—	—	H~R	—	—
MLLM+SoM+AFKG +TETR	—	—	—	H~R	—	—	—	—	—	—	—	—
ControlMLLM+AFKG +TETR	—	—	—	H~R	R~H	—	—	—	—	—	—	—
Mark-Aware+MLLM+AFKG +TETR	—	—	—	**H~R**	**R~H**	—	—	—	—	—	**R~H**	—

(—: no changes; H~R: reallocate to robot; R~H: reallocate to human)

1) To achieve task rescheduling for dynamic human-robot collaboration in disassembly tasks, this paper proposes a human-robot collaboration task rescheduling framework empowered by MLLM and KG. It facilitates scene perception on previously unseen product models, matching them with similar disassembly scenarios for human-robot task allocation.
2) To address the issue of human intent recognition without human physical actions and the understanding of complex spatial scenarios in professional domains, this paper proposes a Mark-Aware prompted MLLM method. This method combines mixed reality (MR) eye-tracking technology, enabling interaction with the model through gaze points under action-free conditions, achieving accurate identification of constraint relationships within complex spatial relations.
3) To address the challenge of recognizing previously unseen parts in dynamic disassembly scenarios, this paper introduces the concept of disassembly affordance and constructs an Affordance Knowledge Graph (AFKG). This method effectively integrates geometric features of appearances within the same disassembly action scenario and achieves precise alignment between LLM information and KG nodes, thereby enhancing the accuracy and practicality of MLLM + KG in identifying unseen parts.

However, some issues still need to be addressed: (1) This paper focuses solely on the robot's dynamic scene perception capabilities and its operational constraints, without addressing the operational requirements of robots in dynamic scenarios. Future research could further explore issues related to robot operations in dynamic environments. (2) This paper only considers single human single robot collaboration. Future research could explore the domain of multi-human and multi-robot collaboration.

CRediT authorship contribution statement

Yu Weigang: Writing – original draft, Methodology, Investigation. **Lv Jianhao:** Methodology, Investigation, Formal analysis. **Zhuang Weibin:** Validation, Resources, Methodology. **Pan Xinyu:** Software, Data curation. **Wen Sijie:** Visualization, Conceptualization. **Bao Jinsong:** Resources, Project administration, Funding acquisition. **Li Xinyu:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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